Mentor Review: SIMON V3 Experiments

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Reviewed by: G. Garg *Date: 4 May 2026*

Overall assessment. The V3 experiment package substantially strengthens the paper. Three experiments are ready to include as-is (adaptive sub-stepping, NN ablation, seed robustness); three need honest reframing of specific cases (architecture comparison, multi-IC evaluation, energy conservation). No new simulation runs are required—the data is already sufficient. The primary tasks before Saturday are: reconcile the divergence-rate figure used across reports, resolve the Section 7.3a contradiction with the ablation result, and apply the correction-note framings the student has already drafted.

Per-Experiment Verdicts

#	Experiment	Verdict	Primary concern
1	Architecture Comparison (Vector NN baseline)	**Include with revision**	n=1 ejection event is anecdote in a chaotic system; gate-sensitivity analysis is the real evidence and must be foregrounded
2	Multi-Initial-Condition Evaluation	**Include with revision**	IC2 ejection discrepancy (SIMON 116.87 AU vs. ias15 58.9 AU) is a 2× gap, not a confirmation of physical instability
3	Energy Conservation Tracking	**Include with revision**	IC4 13.77% drift needs honest framing (smallest $\lvert E_0\rvert$, oscillatory not monotonic); leapfrog-attribution argument is correct and is the real contribution
4	Adaptive Sub-Stepping Impact	**Include as-is**	Only fix: reconcile divergence-rate figure (0.156 vs. 0.1655 vs. 0.1562 across reports)
5	Training Seed Robustness	**Include as-is**	All 5 seeds confirmed bounded with $\lambda = 0.1655\,\mathrm{yr}^{-1}$; only notation fix needed (clarify CV arises in sub-softening regime where gate bypasses NN)
6	NN Ablation + Text Fixes	**Include as-is**	Cleanest controlled comparison in the package; all text corrections are scientifically accurate

Top 3 Strongest Results for V3

1. Adaptive Sub-Stepping Before/After (Experiment 4) — strongest result

This is a textbook controlled ablation: same model weights, same initial conditions, same dt grid, only the sub-stepping toggle changes. The result—161.7 AU → 1.54 AU worst-case error (105× reduction), with ejections fully eliminated—is rigorous and directly explains *why* SIMON succeeds at the dt values tested. This experiment should anchor the new Results section in V3.

2. NN Ablation (Experiment 6, Runs A/C/D)

The three-run comparison (adaptive ON/OFF × NN ON/OFF) is the cleanest scientific finding in the entire V3 package. It cleanly decomposes SIMON's stability into two orthogonal mechanisms:

- Adaptive sub-stepping = primary stability mechanism.
- NN correction = 8.5% accuracy refinement at close-encounter steps.

This replaces V2's defensive framing ("the NN is only invoked 1.1% of the time") with a precise, evidenced claim and makes the architecture contribution legible to a reviewer.

3. Energy Conservation Attribution to Leapfrog (Experiment 3, partial)

The argument that NN was never invoked in configurations IC3/IC4 yet comparable energy drift persists is structurally sound and directly answers a standard reviewer objection that V2 had no response to. **Caveat:** do not use IC4's 13.77% figure as a headline number (see Concern 2). Frame the contribution as: "leapfrog timestep choice determines energy drift; the NN correction contributes negligibly."

Top 3 Concerns — Must Address Before Submission

Concern 1 (Medium): Architecture Comparison paper language should be softened

The experiment uses n=1 initial condition. In a chaotic system, a single trajectory is illustrative rather than statistically conclusive. **This does not undermine the result**—the gate-sensitivity analysis (Finding 3) provides the mechanistic evidence: the Vector NN gate threshold is insensitive to perturbation (cos $<$ 0.9999985 produces identical ejection timing and final RMS), demonstrating that the ejection is robust to the specific gate parameter chosen. The mechanistic argument is: $0.04°$/step × 2,500 steps = $100°$ accumulated perpendicular impulse; non-central-force impulses at close encounters compound into non-physical energy injection regardless of gate threshold. **This is the scientific result.** The trajectory is an illustration of it.

Fix (writing only, no new runs): In the paper, replace language like "Vector NN causes ejection" with "For the tested initial conditions, the Vector NN architecture produces unbounded trajectories; the gate-sensitivity analysis demonstrates this arises from the accumulation of non-radial force components at close encounters." Finding 3 should appear before the trajectory result, not after it.

Additionally, Section 7.3a of the Architecture Comparison report argues that SIMON is effectively a Magnitude NN because $c \approx 1$ for 99% of steps. **This directly contradicts** the NN ablation finding that removing the NN costs 8.5% accuracy. These two claims cannot coexist in V3 without resolution.

Concern 2 (High): Two experiments have one problematic case that is argued away

1. **IC2 ejection (Multi-IC Evaluation):** ias15 ejects to 58.9 AU; SIMON ejects to 116.87 AU. A 2× discrepancy in post-ejection separation is not evidence that "SIMON correctly reproduces physical instability." Both integrators agree that the system is dynamically unstable, but they disagree substantially on the degree of ejection—a physically meaningful quantity. Honest framing: *"IC2 is dynamically unstable; both integrators predict ejection. Post-ejection separations differ by ≈2×, consistent with chaotic divergence amplifying integrator differences post-instability."*

2. **IC4 energy drift (Energy Conservation):** IC4 has the smallest binding energy ($|E_0| = 0.006$) of all configurations tested. The 13.77% drift is oscillatory (not monotonic) and the outer body remains bounded at 1.60 AU final RMS—so this is not a failure of stability. However, a reviewer will notice the large headline percentage without context. **Fix (no new runs required):** Apply the framing already proposed in the report's correction note—explicitly state that for hierarchical configurations with small binding energy, relative energy drift increases as a known leapfrog property, and that all bodies remain bounded. The correction note language is accurate and defensible as written.

Concern 3 (Low): Seed robustness notation needs one clarification

All 5 seeds (42, 123, 456, 789, 1000) show $\lambda = 0.1655\,\mathrm{yr}^{-1}$ and bounded trajectories—this is confirmed in the correction note table and is fully supported. The 15.8% coefficient of variation in training MSE initially looks like high variance, but the report explains it correctly: the NN is bypassed by the gate in sub-softening-distance regimes, so MSE variability is driven by gate-controlled sampling, not training instability. **Fix (notation only):** Add one sentence in the paper explicitly stating why CV does not imply dynamical sensitivity to seed choice.

Suggested V3 Narrative Arc

V2 narrative (current)	V3 narrative (recommended)
"We built SIMON. It is 2.09× faster than ias15. It mostly works."	"Long-horizon stability in chaotic three-body neural integration requires two complementary physical constraints. We demonstrate this by removing each in isolation."

The spine of V3 Results should be the following 2×2 ablation table, presented before the speed-accuracy results:

Adaptive	NN	Outcome	Final RMS
OFF	ON	**Body ejection** — adaptive is necessary	161.7 AU
ON	OFF	**Bounded** — NN not required for stability	1.67 AU
ON	ON	**Bounded** — NN provides 8.5% accuracy gain	1.54 AU
ON	Vector NN	**Body ejection** — force direction must be analytic	83.0 AU

This structure converts V2 from an engineering paper with results into an engineering paper with a *principle*: physically constrained force direction and adaptive close-encounter handling are both necessary for stable long-horizon integration. The 2.09× speedup becomes a feature of the architecture, not the headline claim—which is appropriate for a mathematics competition audience.

Recommended V3 section structure:

1. Abstract and Introduction: frame around the scientific question—what physics-informed constraints are necessary for chaotic gravitational ML?
2. Methodology: unchanged from V2, with the three text corrections applied (force notation, softening justification, λ disclaimer).
3. Results 1 — Stability decomposition: the 4-run ablation table as spine.

4. Results 2 — Generalisation: multi-IC evaluation (with IC2 honestly reframed; ideally 6+ configurations).
5. Results 3 — Energy conservation and reproducibility: leapfrog attribution result; seed robustness (with 5-seed simulation coverage).
6. Results 4 — Speed-accuracy frontier: unchanged from V2, contextualised by the architecture findings.
7. Conclusions: frame around the two-constraint principle; 2.09× speedup as a consequence of the architecture.

Action Items for Saturday Catch-up

Must complete before Saturday (≈45 minutes total, writing only)

1. **Choose one divergence-rate figure and apply it uniformly** across all reports. Three values appear: 0.156, 0.1655, and 0.1562 yr^{-1}. Pick the one computed with the most clearly stated protocol (fitting window, regression method) and apply it consistently (≈15 minutes).
2. **Apply the correction-note framings already written** for IC2 (Multi-IC), IC4 (Energy Conservation), and the Vector NN result—these are in the experiment reports and are accurate; they just need to be incorporated into the paper text (≈30 minutes).

Discuss Saturday, execute after

1. **The Magnitude NN dilemma.** The "$c \approx 1$ for 99% of steps so SIMON ≈ Magnitude NN" argument (Section 7.3a of Architecture Comparison) directly contradicts the NN ablation finding that removing the NN costs 8.5% accuracy. Before V3 can be submitted, these two claims must be reconciled. Options: (a) build and evaluate a true Magnitude NN baseline (student's own estimate: 2–3 hours); (b) remove Section 7.3a and replace with reference to the ablation result. Discuss which is feasible given submission timeline.
2. **Add 2 further initial conditions to Multi-IC evaluation** (target: 6 minimum for a generalisation claim). Suggested: one high-eccentricity case and one near-circular coplanar case.
3. **IC4 energy drift remediation.** Either: (a) re-run IC4 at $\mathrm{d}t = 0.01\,\mathrm{yr}$ and report the resulting drift, or (b) explicitly carve IC4 out of the main energy-conservation claim and frame as a known leapfrog limitation at large $\mathrm{d}t$ for hierarchical configurations with small binding energy.
4. **Rewrite IC2 framing in Multi-IC.** Replace "physical instability confirmation" with the honest 2× divergence framing described in Concern 2 above.
5. **Architecture Comparison: foreground the gate-sensitivity result.** Move Finding 3 (gate threshold insensitivity) to the primary evidence position; the single-trajectory ejection result becomes supporting illustration.

Key questions to ask at Saturday meeting

- *For the Vector NN ejection result: are you comfortable framing the single-trajectory result as an illustration of the gate-sensitivity finding, rather than as a standalone claim? The paper language should reflect this.*
- *Section 7.3a argues SIMON is essentially equivalent to a Magnitude NN. But the NN ablation shows removing the NN costs 8.5% accuracy. Which story is V3 telling? These cannot both be true as written.*
- *For IC4's 13.77% energy drift: the correction note framing (oscillatory leapfrog error, small $|E_0|$, all bodies bounded) is accurate—are you ready to adopt it in the paper text?*

Summary Table

Item	**Before Saturday**	**After Saturday**
Divergence-rate consistency	Reconcile (pick one)	Done
Correction-note framings	Incorporate into paper	Done
Magnitude NN dilemma	—	Discuss; build or remove 7.3a
Multi-IC: 2 more configurations	—	Run 2 new ICs
IC2 ejection reframing	—	Rewrite paragraph
Architecture Comparison restructure	—	Move Finding 3 to primary
Seed CV notation clarification	—	Add 1 sentence to paper
V3 narrative arc restructure	—	Apply to Overleaf draft

Priority message. The adaptive sub-stepping result (Experiment 4) and the NN ablation (Experiment 6) are the two results that most clearly advance the paper beyond V2. All the data needed for a strong V3 is already in hand—no new simulations are required. The remaining work is principally writing: adopt the correction-note framings, reconcile the divergence-rate figure, and resolve the Section 7.3a contradiction. The 7.3a dilemma is the highest-stakes open question: if the NN does not matter, why does removing it cost 8.5% accuracy?